Project: A1 Credit risk

Consultants: [FLIORIS Vassily] – [r1142362]

Part 1: Data exploration and visualization

Replace the prompts with your answers. Keep the report to three pages.

Variables and grouping choices

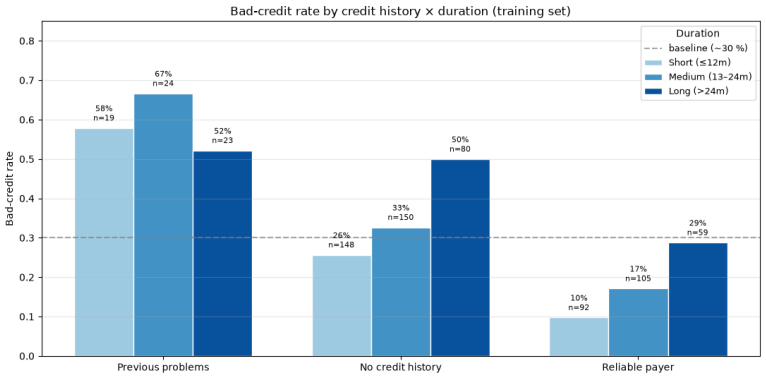
After computing spread and monotonicity on all 20 pred (see graph 1.3.2-EDA) I chose :
1. credit_history : top of the spread ranks, and nice montone. Also past behaviour of past credit history is logically important aswell.
2. duration : the best quantitative pred, also longer loans can also mean more time for problems.

→ *Status* was also considered (cleaner symmetric shape, better spread than duration, monotone, see 1.3.2 graph) but the assymetry of *credit_history* I thaught was more interesting to look at, and I chose *duration* also instead, because pairing *credit_history* with *status* would show the same categorical pattern.

Grouping choise :
- **credit_history :** had 5 categories, I grouped 0&1 to mean "previous problems", 2 as "no-credit history" and grouped 3&4 to mean "good history". This fits our bad/good creditor bussiness meaning.
- **duration :** simply rebinned into 3 ranges (0-12, 13-24 and 24+) instead of the original five, which was noise in the middle (see 1.3.2 tables). Now its better monotone.

Figure

Show bad-credit rates and sample sizes for the joint groups. Use descriptive category labels.



	History	Duration	n	Bad rate	Bad credits
0	Previous problems	Short (≤12m)	19	57.9 %	11
1	Previous problems	Medium (13–24m)	24	66.7 %	16
2	Previous problems	Long (>24m)	23	52.2 %	12
3	No credit history	Short (≤12m)	148	25.7 %	38
4	No credit history	Medium (13–24m)	150	32.7 %	49
5	No credit history	Long (>24m)	80	50.0 %	40
6	Reliable payer	Short (≤12m)	92	9.8 %	9
7	Reliable payer	Medium (13–24m)	105	17.1 %	18
8	Reliable payer	Long (>24m)	59	28.8 %	17

Table above is values of figure.

Comments:

	Pick	History	Duration	n	Bad credits	Bad rate
0	Highest bad-rate	Previous problems	Medium (13–24m)	24	16	66.7%
1	Largest bad-credit volume	No credit history	Medium (13–24m)	150	49	32.7%

Table showing the group with the highest bad-rate (A) and the largest bad-credit (B).

- **(A) Highest bad-rate :** x2 the bad rate of B, but with only n=24, so can be very noisy if we add new candidates the bad-rate change a lot, not very reliable.
- **(B) Largest bad-credit volume :** half the bad-rate of A but x6 bigger population, so way more reliable.
- **Reliability :** The merge of *history* combined different bad rates (which add noise but make the cells less tiny and more bussiness meaning, so in fine the groupings give more reliability; same for *duration*).
- **Bank :** use A to target high-risk applicants, but the larger expected loss is in group B. If reviews are cheap and bad loans are expensive, targeting B avoids 49 bads vs 16.

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Part 2: WOE and logistic regression

Predictor selection and level choices

Started by re-grouping all 20 predictors and computing their WoE and IV, (binning strategy following Dimitri Bianco's method) + following business meaning, see the notebook for actual per-variable details :

	variable	iv	strength	levels
0	status	0.650	suspicious	No account | Overdrawn | Low balance (0–200) | Healthy (≥200)
1	credit_history	0.395	strong	Previous problems | Stable / no history | Reliable payer
2	duration	0.243	medium	Long (>30m) | Medium (16–30m) | Short (9–15m) | Very short (≤8m)
3	savings	0.206	medium	No / small savings | Real savings
4	property	0.132	medium	Real estate | Car / insurance | No property
5	employment_duration	0.106	medium	Unstable | Medium 1–4y | Stable ≥4y

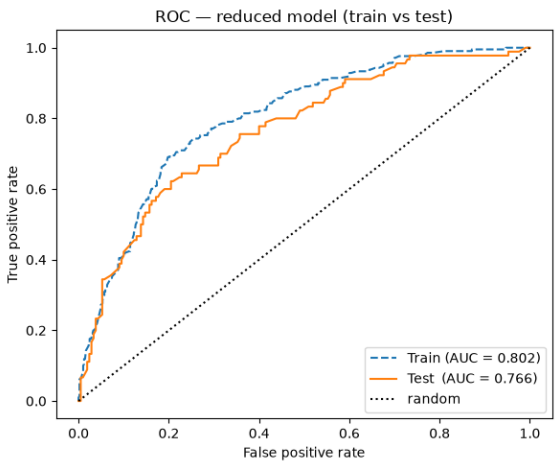
1. merge bins with n<35 with the closest WoE-noeighbor.
2. merge zigzags, when WoE goes up and down by more than 0.15, the middle bin is noise, merge it with its closest WoE.
3. aim for to keep 2-4 bins minimum.
4. Increase for monotone WoE
5. drop variables with IV < 0.02

Then from this resulting table, I created a few datasets (**"full dataset"** -all 8 preds, "**reduced dataset**" - dropped weak preds, "**minimal dataset**" – only *status*, to measure its effect).

Also considered adding an ethical dataset to measure the impact of predictors : *personal_status_sex* (IV 0.076) and *foreign_worker* (IV 0.048) for a quick bias check. They were dropped for those ethical reasons, (fortunatley both had weak IV, so nothing predictive was really lost).
To go further (but out of scope), create this ethical dataset and measure the impact.

→ Final chosen dataset is "**reduced-dataset**" bexause it had the same AUC as **full-dataset**, but with less predictors 6 vs. 8, so we get the same performances for less features (see notebook).

Figure – ROC curve of the chosen dataset (reduced, 6 predictors)



- **Highest-IV predictor :** status (IV 0.65). very high IV value, above 0.5 can usually mean leakage, but after checking the bad rates for each of its levels, it is nice and monotone with large cells. So no leakage , status is simply a very strong signale.

- **WoE :** Riskiest=No account (bads are x2.3 more than goods, 49% bad rate). Safest=Healthy (goods are x3.1 more than bads, only 12% defaulted).

- **No overfitting**, because : 1. Gap between Train and Test AUC is very small, 2. model with very few params (6) for 700 training rows. 3. WoE binning I used forces 35+ rows per level, so no small cells. All these points make overfitting very unlikely. Perfomance is moderate, better than random but prob not enough for real world.

- **Conclusion** : bank can use model to rank applicants (who to review, who to approve) but not to predict, we need to calibrate the model, predictions are not guaranteed to be correct, th bank cannot use it for example to decide what interest rate to charge. Only to separate good applicants from bad ones. Another limitation is that the dataset is very small, and its data comes only from **one** bank a long time ago, not likely that it can predict for today.

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Part 3: Scorecard

Full scorecard

	predictor	level	WOE	points	swing
0	status	No account	-0.8162	66.21	45.38
1	status	Overdrawn	-0.4016	75.89	
2	status	Low balance (0–200)	0.5698	98.56	
3	status	Healthy (≥200)	1.1276	111.59	
4	credit_history	Previous problems	-1.2150	58.27	49.3
5	credit_history	Stable / no history	-0.1438	82.07	
6	credit_history	Reliable payer	1.0044	107.57	
7	duration	Long (>30m)	-0.7333	66.63	49.95
8	duration	Medium (16–30m)	-0.0805	83.22	
9	duration	Short (9–15m)	0.2569	91.79	
10	duration	Very short (≤8m)	1.2321	116.58	
11	savings	No / small savings	-0.2652	79.30	23.75
12	savings	Real savings	0.7911	103.05	
13	property	Real estate	-0.6120	72.17	23.85
14	property	Car / insurance	-0.0429	84.35	
15	property	No property	0.5026	96.02	
16	employment_duration	Unstable	-0.4315	74.04	20.57
17	employment_duration	Medium 1–4y	-0.1338	81.78	
18	employment_duration	Stable ≥4y	0.3594	94.61	

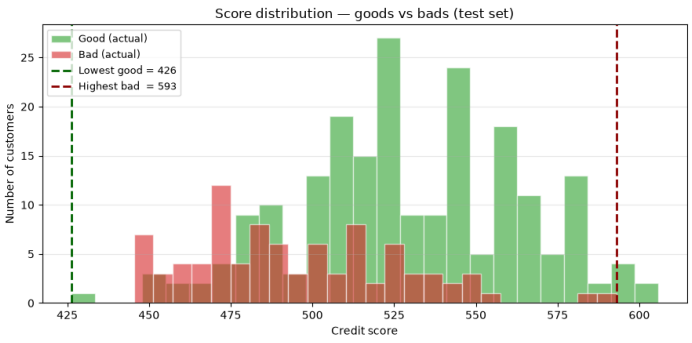
Notes :
- WoE and points move together, good sign.
- at WoE=0, points=85 (concord with the base slice computed from the formula).

- in the notebook, I checked that the scale specifications (score 600 ↔ odds 50:1, and every +20 points doubles the odds) are verified on the test data aswell. Another check is made for the scorecard sum to match the model score to 0.000000 across all test rows(300).

- I also compute the Swing : how much points one variable different options can affect the final score.

- The top 3 swing are x2 the bottom ones, our scorecard has only 6 variables, so those 3 features dominate the prediction. Same top-3 as the top IV ranking in Part 2 (*status*, *credit_history*, *duration*). An applicant's total score is mostly decided by these three variables.

Figure – final test scores boxplot/histogram



Notes :
- I found that a histogram is more interesting to showcase (boxplot is in notebook aswel). Here we can also see how both distribution overlap at (470-560), which is exactly the AUC 0.766 error zone (~people the model cannot distinguish good from bad). We can also see the group density better.

- The two dashed lines are the extreme errors. The lowest good-applicant is rated riskier than all 90 defaulters, while the best-bad sits in the higher percentiles of the good distribution.

- If the bank used this model to rank applicants, it would have rejected a good candidate and validated a bad one. Even if our model (AUC 0.766) is not amazing and we could improve it to AUC 0.95, no classifier can ever be perfect, so the bank has to accept a certain error rate.

If the bank wants to use this model do it like this :
1. Score > 560 : approve (very few bads in this region).
2. Score < 470 : Reject or review manually. (mostly bads in region).
3. in between : Review manually (both groups presenet here.)

Do not use for accurate prediction before calibrating the model.